Review Comments – Reviewer 1

1. Dataset section discusses both the 5,856-image Kaggle dataset and NIH ChestX-ray14 without clearly stating whether NIH data are actually used for training or testing, so the exact dataset composition and source of every experimental sample should be clarified.

   Response: We sincerely thank the reviewer for this valuable comment. In accordance with the reviewer's suggestion, we have clarified the relationship between the Kaggle dataset and the NIH ChestX-ray14 dataset in the revised manuscript. Both datasets were initially studied and evaluated for their suitability for pneumonia detection. However, the NIH ChestX-ray14 dataset is very large, requiring substantial computational resources for image processing and model training. Due to the limited availability of high-performance computing resources and GPU memory, it was not feasible to perform the complete training and testing process using the full NIH ChestX-ray14 dataset within the scope of the present study.

- Therefore, after evaluating both datasets, the Kaggle dataset was selected for the final experimental implementation and model training/testing because it was more computationally manageable and suitable for the available hardware resources. The Kaggle dataset used in this study has been appropriately cited as Reference [22] in the revised manuscript. We have also revised the dataset description to clearly distinguish the two datasets and explain the rationale behind selecting the Kaggle dataset for the final experiments.

- The Pneumonia simple's collection from Kaggle has 5,856 CXR that are divided into two primary categories: plain and Pneumonia images [22]. This dataset was specifically chosen because of its clinical reliability and balanced representation of paediatric chest X-rays, making it suitable for benchmarking deep learning-based diagnostic models. Each image in the dataset was resized to a uniform resolution of 224 × 224 pixels and normalized to the [0,1] range to ensure computational consistency and prevent pixel-scale bias during model training. The images were converted into three-channel RGB format to align with Efficient Net's input requirements [22] [24].

2. Kaggle dataset is described as having a "balanced representation," although this commonly used pediatric pneumonia dataset has unequal Normal and Pneumonia class sizes, so the actual class distribution and any balancing method should be reported correctly.

Response: We sincerely thank the reviewer for identifying this important issue. We acknowledge that describing the Kaggle dataset as having a "balanced representation" was inaccurate because the dataset contains an unequal number of Normal and Pneumonia images. This statement was included inadvertently during the manuscript preparation.In accordance with the reviewer's valuable suggestion, we have removed the term "balanced representation" from the revised manuscript and have clearly reported the actual class distribution of the dataset. We have also clarified the data-processing procedure used in our study and have avoided claiming that class balancing was performed unless explicitly supported by the implemented methodology. The revised dataset description now accurately reflects the original class distribution and the preprocessing steps applied before model training.

3. Data split is given only as a general range of 70–80% training, 10–15% validation, and 10–15% testing instead of the exact split used, so precise sample counts and patient-level separation for each subset should be provided.

   Response: -We sincerely thank the reviewer for this valuable observation. In response to the reviewer's comment, we have revised the Data Segmentation subsection to report the exact dataset split used in our experiments rather than providing general percentage ranges. The dataset was divided into 80% for training, 10% for validation, and 10% for testing.

   Data Segmentation The dataset was divided into three subsets to ensure effective model training, validation, and performance evaluation:

   - *Training Set: Use 80% of the dataset for training purposes.*
   - *Validation Set: Datasets comprising 10% of the validation set.*
   - *Testing Set: 10% of the whole data collection.*

4. Model description is technically inconsistent because the methodology states sigmoid activation for binary classification while the architecture later uses a two-unit Softmax output, so one consistent output layer and corresponding loss function should be clearly specified.

   Response: We sincerely apologize for this inconsistency and thank the reviewer for identifying the error. The statement indicating that **sigmoid activation** was used for binary classification was included inadvertently during the manuscript preparation. In the actual implementation, the proposed model uses a **two-unit Softmax output layer**, as correctly shown in **Figure 3**, to classify the images into the two target classes. Therefore, to maintain consistency between the methodology, model architecture, and experimental implementation, we have corrected the activation function description in the revised manuscript from **sigmoid** to **Softmax**. The corresponding loss function and output-layer description have also been carefully reviewed and revised to accurately reflect the implemented model.

Softmax activation function is used to convert the outputs of the two neurons in the final layer into class probabilities:

$$y_i = \sigma(z_i) = \frac{e^{zi}}{\sum_{j=1}^{2} e^{zi}} \quad i = 1,2 \quad (4)$$

Where y_i is the predicted probability of pneumonia.

5. Training accuracy reaches 99.11% while validation and test accuracies are lower, but no learning curves with proper numerical analysis, early stopping details, or generalization assessment are provided, so possible overfitting should be investigated more carefully.

   Response:-We sincerely thank the reviewer for this valuable observation. In response to the reviewer's suggestion, we have added the **training and validation learning curves for loss and accuracy** to the revised manuscript to provide a clearer assessment of model convergence and possible overfitting. The model was trained for **17 epochs**, and the learning curves show that the training and validation losses consistently decreased during training, while the corresponding accuracies improved substantially after Epoch 12. The highest validation accuracy of **96.44%** was achieved at Epoch 16, whereas the lowest validation loss of **0.3153** was obtained at Epoch 17.

The final model achieved 99.11% training accuracy, 96.43% validation accuracy, and 96.80% test accuracy. The difference between training and validation accuracy was approximately 2.68 percentage points, while the difference between training and test accuracy was approximately 2.31 percentage points. These relatively small generalization gaps indicate that the model does not exhibit severe overfitting. The test-set classification results further demonstrate balanced performance, with an overall accuracy of approximately 96.80% and a weighted F1-score of 0.97.

We have therefore included the learning curves and corresponding numerical analysis in the revised manuscript to provide a more transparent evaluation of model convergence, generalization capability, and potential overfitting.

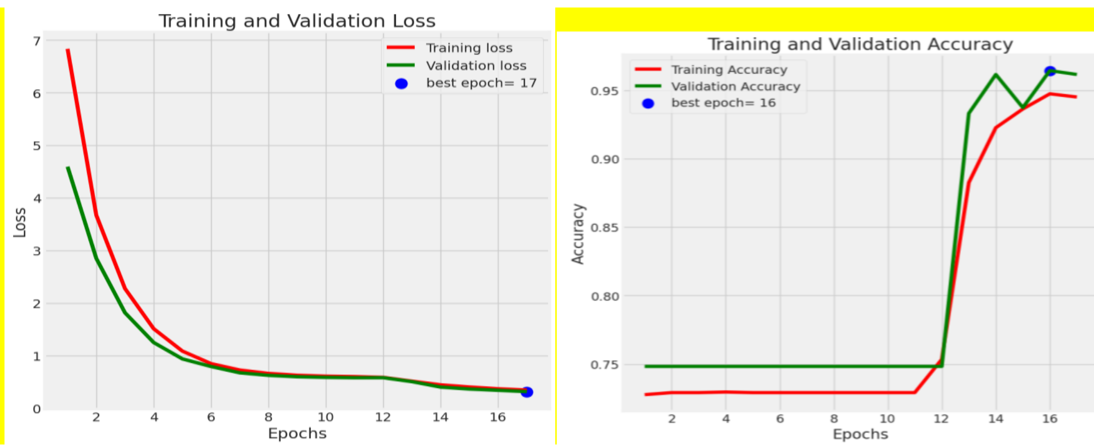

Fig. 5. Graphical representation of training and verification results. Precision and Deviation

6. Figure 4 requires revision, as the overlapping fonts make the figure difficult to interpret. Figures 1, 2, and 4 are not cited in the manuscript. Kindly add the corresponding in-text citations. Redraft the "Conclusions" into a single paragraph. Tables 2 and 3 are not cited in the manuscript. Kindly add the corresponding in-text citations.

   Response: We sincerely thank the reviewer for these valuable suggestions. In accordance with the reviewer's comments, the following revisions have been made in the revised manuscript:Figure 4, Figure Citations, Conclusion and Table Citations These revisions have been carefully incorporated to improve the clarity, readability, organization, and overall presentation of the manuscript.

Review Comments – Reviewer 2

1. Use consistent terminology by replacing “respiratory problem” with “pneumonia” where appropriate throughout the manuscript.

   Response: We sincerely thank the reviewer for this valuable comment. In accordance with the reviewer’s suggestion, we have carefully revised the manuscript and replaced the term **“respiratory problem”** with **“pneumonia”** wherever appropriate to ensure consistent and precise terminology throughout the manuscript. These changes have been incorporated in the relevant sections of the revised manuscript.

2. Kindly correct the model’s name inconsistency where “EfficientNet80” appears instead of “EfficientNetB0” in the abstract.

   Response: We sincerely thank the reviewer for pointing out this inconsistency. As per the reviewer’s valuable suggestion, we have carefully checked the manuscript and corrected the model’s name from **“EfficientNet80”** to **“EfficientNetB0”** in the abstract. The correction has also been verified to ensure consistency with the model name used throughout the manuscript.

   To develop a simplified approach and reduce over-fitting, the proposed EfficientNetB0 employs a pre-trained EfficientNetB0 design with ImageNet weights as an attribute extractor, followed by batch normalization, fully connected layers with L1/L2 regularization, and dropout. Transfer learning is used to capitalize on rich visual representations gained from large-scale image collections

3. Clarify the relationship between the Kaggle dataset and the NIH ChestX-ray14 dataset in the Dataset description.

   Response: We sincerely thank the reviewer for this valuable comment. In accordance with the reviewer’s suggestion, we have clarified the relationship between the Kaggle dataset and the NIH ChestX-ray14 dataset in the revised manuscript. Both datasets were initially studied and evaluated for their suitability for pneumonia detection. However, the NIH ChestX-ray14 dataset is very large, requiring substantial computational resources for image processing and model training. Due to the limited availability of high-performance computing resources and GPU memory, it was not feasible to perform the complete training and testing process using the full NIH ChestX-ray14 dataset within the scope of the present study.

   Therefore, after evaluating both datasets, the Kaggle dataset was selected for the final experimental implementation and model training/testing because it was more computationally manageable and suitable for the available hardware resources. The Kaggle dataset used in this study has been appropriately cited as Reference [22] in the revised manuscript. We have also revised the dataset description to clearly distinguish the two datasets and explain the rationale behind selecting the Kaggle dataset for the final experiments.

The Pneumonia simple's collection from Kaggle has 5,856 CXR that are divided into two primary categories: plain and Pneumonia images [22]. This dataset was specifically chosen because of its clinical reliability and balanced representation of paediatric chest X-rays, making it suitable for benchmarking deep learning-based diagnostic models. Each image in the dataset was resized to a uniform resolution of 224 × 224 pixels and normalized to the [0,1] range to ensure computational consistency and prevent pixel-scale bias during model training. The images were converted into three-channel RGB format to align with Efficient Net's input requirements [22] [24].

4. It is suggested to improve the formatting of the Data Segmentation subsection by presenting the dataset split as a structured list.

   Response-We sincerely thank the reviewer for this valuable suggestion. In accordance with the reviewer's comment, we have revised the Data Segmentation subsection and presented the dataset distribution as a structured list. The dataset is now clearly divided into training (70–80%), validation (10–15%), and testing (10–15%) subsets to improve the clarity, readability, and presentation of the manuscript.

   Data Segmentation

   The dataset was divided into three subsets to ensure effective model training, validation, and performance evaluation:

   - Training Set: Use 70-80% of the dataset for training purposes.
   - Validation Set: Datasets comprising 10-15% of the validation set.
   - Testing Set: Ten to fifteen percent of the whole data collection.

5. Authors should revise the phrase "develop simpFlification" in the abstract because it contains a typographical error.

   Response-We sincerely thank the reviewer for identifying this typographical error. As per the reviewer's valuable suggestion, we have carefully revised the abstract and corrected the phrase **"develop simpFlification"** to the appropriate wording. The revised text has also been thoroughly checked to ensure that similar typographical and grammatical errors are avoided throughout the manuscript.

   To develop a simplified approachand reduce over-fitting, the proposed EfficientNetB0 employs a pre-trained EfficientNetB0 design with ImageNet weights as an attribute extractor, followed by batch normalization, fully connected layers with L1/L2 regularization, and dropout.